EXPERIMENT 5

Bank Loan Application Form (Advanced) — Advanced Fields Use Case

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Subject: Low Code Application Development using Zoho Creator
Project: Bank Loan Application & Pre-Screening Management

Aim

To extend the Loan_Application_Form of Experiment 4 with all seventeen Advanced Fields of the Form Builder, so that the form supports joint applications, branch verification, document and media upload, automatic EMI estimation, and a unique scannable application reference.

Introduction

The basic form captures who the applicant is and what loan is wanted. A real bank also needs to verify the servicing branch against a master list, record co-applicants in the same submission, accept income proof and a property video, compute an estimated instalment, and issue a tracking reference. Zoho Creator's Advanced Fields cover these needs: lookups and subforms for related data, file-type fields for media, and Auto Number, Formula and QR/Barcode fields for values the system generates for itself.

Procedure

1. Create a master form **Branches** with Branch_Name, IFSC_Code and Manager_Name, and add three branch records.
2. Open Loan_Application_Form in the Form Builder and expand the Advanced Fields palette.
3. Add the seventeen fields to the sections of Table 5.2, using the link names given there.
4. For the Lookup, point Servicing_Branch_Lookup at the Branches form and show IFSC_Code and Manager_Name as additional display values.
5. For the Subform, add Co_Applicant_Details with columns Co_Applicant_Name, Relationship and Co_Applicant_Income; set the row limit to 0–3.
6. Configure the formats, sizes and ranges listed in Table 5.3.
7. Create a Formula field for the estimated EMI and an Auto Number field with the prefix **BNK-LN-**, the year and a six-digit sequence; attach a QR/Barcode field to the Auto Number.
8. Submit two test applications (one single, one joint) and open the report to confirm the generated ID, QR code and EMI.

Advanced Fields Used and Their Mapping

Table 5.1: Mapping of the seventeen Advanced Fields (sample values from the assignment specification)

Field type	Field name	Purpose	Sample value / behaviour
Decision Box	Has_Existing_Loan_With_Bank	Yes/No: applicant already holds a loan with the bank	Yes / No toggle
Rich Text	Loan_Terms_Display	Formatted terms and conditions inside the form	“Interest rates are subject to change…”
URL	Property_Listing_URL	Link to the property listing (home loans)	https://realestate.example.com/listing/44219
Image	Applicant_Photo	Passport-size photo for KYC	applicant_photo.jpg
Percent	Interest_Rate_Offered	Rate offered from the credit profile	9.25%
Currency	Loan_Amount_Requested	Amount wanted, shown with currency symbol	INR 15,00,000.00
Decimal	Debt_To_Income_Ratio	Existing debt-to-income ratio	0.32
Date-Time	Application_Submitted_On	Exact submission date and time	Recorded automatically
Subform	Co_Applicant_Details	Repeating rows for joint applicants	Row 1: Meena Suresh – Spouse – 35,000
Lookup	Servicing_Branch_Lookup	Choose servicing branch; auto-fill IFSC and manager	Anna Nagar Branch – BANK0001234
Signature	Applicant_Digital_Signature	Declaration that the information is accurate	Drawn signature
File Upload	Income_Proof_Upload	Salary slips, Form 16 or bank statements	income_proof_bundle.pdf
Audio	Loan_Purpose_Voice_Note	Optional voice note on the loan purpose	voice_note_05.mp3
Video	Property_Video_Walkthrough	Optional property video	property_walkthrough.mp4
Auto Number	Loan_Application_ID	Unique sequential reference	BNK-LN-2026-000459
Formula	Estimated_Monthly_EMI	Computed instalment (read-only)	See Formula note below
QR/Barcode	Application_Tracking_QR_Code	Scannable code of the application ID	QR image of the ID

Extended Form Layout

Table 5.2: Sections added to the form

Section	Fields added
E — Branch Verification & KYC	Servicing_Branch_Lookup, Applicant_Photo, Income_Proof_Upload, Debt_To_Income_Ratio
F — Joint Application & Media	Co_Applicant_Details (Subform), Property_Listing_URL, Property_Video_Walkthrough, Loan_Purpose_Voice_Note
G — Loan Terms & EMI	Has_Existing_Loan_With_Bank, Loan_Amount_Requested, Interest_Rate_Offered, Estimated_Monthly_EMI
H — Declaration & System Values	Loan_Terms_Display, Applicant_Digital_Signature, Application_Submitted_On, Loan_Application_ID, Application_Tracking_QR_Code

Validation and Configuration Rules

Table 5.3: Validation and configuration rules

Field	Rule
Servicing_Branch_Lookup	Linked to the Branches form; users search and select rather than type
Co_Applicant_Details	Minimum 0 rows, maximum 3 rows
Income_Proof_Upload	PDF, JPG, PNG only; maximum 5 MB
Applicant_Photo	JPG and PNG only; maximum 2 MB
Property_Video_Walkthrough	Maximum 60 seconds and 50 MB
Interest_Rate_Offered	Value range 7 to 18
Loan_Amount_Requested	Non-negative values; INR formatting
Estimated_Monthly_EMI	Read-only; recalculated whenever amount or rate changes
Loan_Application_ID	Prefix BNK-LN-, current year, six-digit sequence
Application_Tracking_QR_Code	Generated from Loan_Application_ID on submission
Signature, Decision Box	Mandatory

Note on the EMI Formula

The assignment defines the estimate as **(Loan_Amount_Requested × Interest_Rate_Offered) / (12 × Standard_Tenure_Months)**. This is a simplified indicator, because it ignores compounding. A bank's actual reducing-balance instalment is $P \cdot r \cdot (1+r)^n / ((1+r)^n - 1)$, where r is the monthly rate and n the number of months. For example, for a loan of 15,00,000 at 9.25% per year over 180 months, the reducing-balance EMI works out to about 15,438. In the record the assignment formula is configured as specified, and the difference is noted so that the estimate is not mistaken for the sanctioned instalment.

Output

SCREENSHOT TO ATTACH

Form Builder with all seventeen Advanced Fields visible (Sections E to H)

Fig 5.1 — Extended form in the Form Builder

SCREENSHOT TO ATTACH

Live preview showing the Subform rows and the computed Estimated Monthly EMI

Fig 5.2 — Live preview with Subform and EMI

SCREENSHOT TO ATTACH

Servicing Branch Lookup in action: search, select, and IFSC/manager auto-filled

Fig 5.3 — Branch lookup in action

SCREENSHOT TO ATTACH

Loan_Application_Report with two records showing Loan Application ID and QR code columns

Fig 5.4 — Report with generated IDs and QR codes

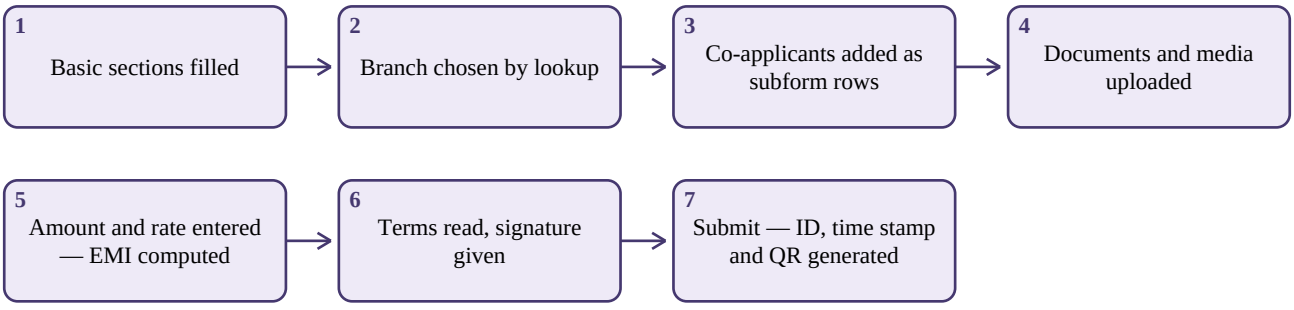


Fig 5.5 — Advanced-field workflow (schematic drawn from the specification; not a screenshot)

Result

All seventeen Advanced Fields were added to Loan_Application_Form with the specified restrictions. The report shows the system-generated application ID, QR code and estimated EMI for each submission.

Conclusion

Advanced Fields move a form from data capture to process support: the Lookup keeps branch data consistent, the Subform handles joint loans in one submission, file fields carry the evidence, and Auto Number, Formula and QR fields let the system produce the identifiers and figures that staff would otherwise create by hand. The EMI formula in the assignment is only an estimate and should be replaced by the reducing-balance formula before any figure is shown to a borrower.